

The level-1 Kashiwara string bound does not lift:
for e -scaled multipartitions at level ℓ the sharp bound is

$$\varepsilon_i \leq \#\{j : s_j \not\equiv i \pmod{e}\}$$

Clio

2026-08-29

Abstract

Let $e \geq 2$, let $\ell \geq 1$, let $\mathbf{s} = (s_1, \dots, s_\ell) \in \mathbb{Z}^\ell$, and let $\mathcal{F}_q[\mathbf{s}]$ be the level- ℓ Uglov q -Fock space of $U_q(\widehat{\mathfrak{sl}}_e)$ with its Kashiwara crystal $(\mathcal{L}, \{[\mu]\}, \tilde{e}_i, \tilde{f}_i)$ on ℓ -multipartitions. For an ℓ -multipartition λ write $e\lambda := (e\lambda^{(1)}, \dots, e\lambda^{(\ell)})$, the multipartition all of whose parts are e times a part of λ .

At level 1 it was proved (2026-08-11) that $\varepsilon_i(e\lambda) \leq 1$ for every partition λ and every i , and this was the engine of the theorem “C5 at $\ell = 1$ ”. The stated expectation for the higher-level lift — that the bound is level-independent — is **false**. We prove:

- (T1) (*Component word.*) For each component j the i -signature of $e\lambda^{(j)}$ at charge s_j , read in increasing content order, is $(\text{AR})^{n_j} \mathbf{A}^{\delta_j}$ with $\delta_j = 1$ iff $s_j \equiv i \pmod{e}$. The level-1 block analysis lifts verbatim, charge-shifted.
- (T2) (*Sharp upper bound, unconditional.*)

$$\varepsilon_i(e\lambda) \leq \#\{j : s_j \not\equiv i \pmod{e} \text{ and } \lambda^{(j)} \neq \emptyset\} \leq \min(|\lambda|, m_i(\mathbf{s})),$$

where $m_i(\mathbf{s}) := \#\{j : s_j \not\equiv i \pmod{e}\} \leq \ell$. The bound holds for *every* tie-breaking convention on equal contents, hence is independent of that choice.

- (T3) (*Sharpness.*) For every $e, \ell, \mathbf{s}, i$ with $m_i(\mathbf{s}) \geq 1$ there is an explicit λ with $\varepsilon_i(e\lambda) = m_i(\mathbf{s})$.

The smallest counterexample to the level-independent bound is $e = 2$, $\ell = 2$, $\mathbf{s} = (0, 0)$, $i = 1$, $\lambda = ((1), (1))$, where $e\lambda = ((2), (2))$ has i -signature AARR and $\varepsilon_1 = 2$, realised by the crystal string $((2), (2)) \rightarrow ((1), (2)) \rightarrow ((1), (1))$.

Consequently the mechanism that proves C5 at level 1 degrades *linearly in the level*: the level-1 “running-max stays ≤ 1 ” heuristic fails because ε_i of a merge of ℓ words is a *sum* of per-component suffix sums, and the corresponding intervals can always be aligned.

We also record what did *not* close. Whether the bound transfers from $e\lambda$ to the Heisenberg descendant $v_{k'}^{[\ell]} = B_{-1}^{k'}|\emptyset; \mathbf{s}\rangle$ depends on a support hypothesis (H2) which we were unable to test: our implementation of the level- ℓ q -wedge straightening rule reproduces all four of Iijima’s worked examples and the entire level-1 theory exactly (79 partitions, 0 mismatches), yet is *not confluent* for $\ell \geq 2$, and no reading inside a searched family of 98,304 variants is both example-consistent and confluent. This is stated precisely in §6 as a located, reproducible blocker.

Total computational verification: 239,693 component-word checks, 1,014,696 bound checks, 66,464 sharpness-construction checks, 497,814 checks of a transposed variant — all reported below, 0 failures against the statements as given.

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1 Setup and conventions

1.1 Multipartitions, contents, residues

Fix $e \geq 2$, $\ell \geq 1$ and a multicharge $\mathbf{s} = (s_1, \dots, s_\ell) \in \mathbb{Z}^\ell$. An ℓ -*multipartition* is a tuple $\boldsymbol{\mu} = (\mu^{(1)}, \dots, \mu^{(\ell)})$ of partitions. A *node* of $\boldsymbol{\mu}$ is a triple (r, c, j) with $0 \leq r$, $0 \leq c < \mu_r^{(j)}$ (0-indexed rows and columns). We extend the notion to *addable* and *removable* nodes in the usual way: writing $L_j := \ell(\mu^{(j)})$ and $\mu_r^{(j)} := 0$ for $r \geq L_j$,

$$\begin{aligned}
(r, \mu_r^{(j)}, j) \text{ is addable} &\iff r = 0 \text{ or } \mu_{r-1}^{(j)} > \mu_r^{(j)} & (0 \leq r \leq L_j), \\
(r, \mu_r^{(j)} - 1, j) \text{ is removable} &\iff r = L_j - 1 \text{ or } \mu_{r+1}^{(j)} < \mu_r^{(j)} & (0 \leq r \leq L_j - 1).
\end{aligned}$$

The *content* of a node is the *integer*

$$\text{ct}(r, c, j) := c - r + s_j \in \mathbb{Z}, \quad \text{res}(r, c, j) := \text{ct}(r, c, j) \bmod e \in \mathbb{Z}/e\mathbb{Z}.$$

A node is an *i-node* if $\text{res} = i$.

1.2 The crystal and ε_i

Let $\mathcal{F}_q[\mathbf{s}]$ be the level- ℓ Uglov q -Fock space [2, 3], a $\mathbb{Q}(q)$ -vector space with standard basis $\{|\boldsymbol{\mu}; \mathbf{s}\rangle\}$ indexed by ℓ -multipartitions. It carries an action of $U_q(\widehat{\mathfrak{sl}}_e)$ and a Kashiwara crystal $(\mathcal{L}, \{[\boldsymbol{\mu}]\})$ with $\mathcal{L} = \bigoplus_{\boldsymbol{\mu}} \mathbb{Z}_{(q)} |\boldsymbol{\mu}; \mathbf{s}\rangle$; the Kashiwara operators $\tilde{e}_i, \tilde{f}_i$ on $\mathcal{L}/q\mathcal{L}$ are computed by the JMМО/FLOTW good-node rule, which we now fix in the exact form used throughout.

Definition 1 (*i*-signature; reading order). Fix $i \in \mathbb{Z}/e\mathbb{Z}$ and a multipartition $\boldsymbol{\mu}$. List all addable and removable *i*-nodes of $\boldsymbol{\mu}$ in **increasing content order**, breaking ties between nodes of equal content by any fixed total order τ on $\{1, \dots, \ell\}$ (nodes of equal content in the same component cannot occur — Lemma 6). Record A for addable and R for removable. The resulting word $w_i(\boldsymbol{\mu})$ is the *i*-signature. Reduce $w_i(\boldsymbol{\mu})$ by repeatedly deleting a factor RA (equivalently: scan left to right

on a stack; an A whose predecessor on the stack is an R cancels it, otherwise push). The reduced word has the form $A^a R^b$; we set $\varepsilon_i(\boldsymbol{\mu}) := b$ and $\varphi_i(\boldsymbol{\mu}) := a$. The operator $\tilde{e}_i$ removes the node of the leftmost surviving R (and is 0 if $b = 0$); $\tilde{e}_i, \tilde{f}_i$ are then extended $\mathbb{Z}$ -linearly to $\mathcal{L}/q\mathcal{L}$, and for $v \in \mathcal{L}$, $\varepsilon_i(v) := \max\{n \geq 0 : \tilde{e}_i^n[v] \neq 0\}$.

Remark 2 (Consistency with the level-1 file). *At $\ell = 1$, $\mathbf{s} = (0)$, an addable node in row r has content $\mu_r - r$, which is strictly decreasing in r ; hence increasing content order coincides with the bottom-up row order used in [5]. The implementation was checked against that paper's worked example ($e = 2$, $\mu = (10, 6, 4, 2)$, $i = 1$: signature ARAR, $\varepsilon_1 = 1$) and against two elementary cases ($\mu = (1)$, $e = 2$: $\varepsilon_1 = 0$, $\varepsilon_0 = 1$). All agree.*

Remark 3 (Which convention is load-bearing). *Exactly one convention is used in the proofs below: the reading order refines increasing content within each component. Everything else — the tie-break τ between components, and any normalisation of the multicharge — is free. Two consequences worth stating, because the multicharge normalisation is a known live ambiguity in this literature (FLOTW normalise $\Delta(\mathbf{s}) = 0$ by restricting $\mathbf{s}$, whereas Uglov does not; the difference is invisible at $\ell = 1$ and load-bearing at $\ell \geq 2$):*

1. *Theorems 9 and 12 are stated for all $\mathbf{s} \in \mathbb{Z}^\ell$, so they specialise to any restricted set of admissible multicharges.*
2. *The counterexample of Example 11 lives at $\mathbf{s} = (0, 0)$, and more generally Corollary 13 applies at $\mathbf{s} = (0, \dots, 0)$ for every $i \neq 0$. So the refutation does not depend on an exotic multicharge and survives every normalisation convention we are aware of.*

The direction of the reading order does matter and is not a free choice: with the order reversed, the same reduction computes φ_i rather than ε_i . Our direction is pinned by three independent checks (Remark 2), of which the sharpest is $e = 2$, $\mu = (1)$: $\varepsilon_0 = 1$, $\varepsilon_1 = 0$, correct because (1) has a removable 0-node and no removable 1-node.

We emphasise that the tie-break τ is left free. Different choices of τ genuinely give different values of ε_i (this occurs in 9,802 of 87,844 sampled instances, see §8), so a theorem valid for all τ is strictly stronger than one valid for a single convention. All our bounds are of this stronger kind; only Definition 1's requirement that the order *refine within-component content order* is used.

1.3 The reduction as a suffix maximum

Lemma 4 (Bracket lemma). *For a word w in $\{A, R\}$ let $\varepsilon(w)$ be the number of R's surviving the reduction of Definition 1. Give R the value +1 and A the value -1, and for a word u let $\|u\|$ be the sum of the values of its letters. Then*

$$\varepsilon(w) = \max\{\|u\| : u \text{ a suffix of } w\}$$

(the maximum includes the empty suffix, so it is ≥ 0).

Proof. Write $M(w)$ for the right-hand side. Induct on $|w|$; both sides are 0 for w empty. Suffixes of $w'R$ are the empty word together with $u'R$ for u' a suffix of w' , and $\|u'R\| = \|u'\| + 1$; since $M(w') \geq 0$ we get $M(w'R) = M(w') + 1$. In the reduction, a final R is pushed and never cancelled by a later letter (there is none), so $\varepsilon(w'R) = \varepsilon(w') + 1$. Similarly $M(w'A) = \max(0, M(w') - 1)$, and in the reduction a final A cancels a surviving R if one exists, so $\varepsilon(w'A) = \max(0, \varepsilon(w') - 1)$. The claim follows by induction. $\square$

2 The component word

Throughout this section fix one component: a partition λ with $L := \ell(\lambda)$, a charge $s \in \mathbb{Z}$, a residue $i \in \mathbb{Z}/e\mathbb{Z}$, and set $\mu := e\lambda$ (all parts multiplied by e). Put

$$b := (s - i) \bmod e \in \{0, 1, \dots, e - 1\}.$$

Lemma 5 (Row residues). *In $\mu = e\lambda$ with charge s : the addable node in row r (present iff $r = 0$ or $\lambda_{r-1} > \lambda_r$, for $0 \leq r \leq L$) has content $e\lambda_r - r + s$, hence residue i iff $r \equiv b \pmod{e}$; and the removable node in row r (present iff $r = L - 1$ or $\lambda_{r+1} < \lambda_r$, for $0 \leq r \leq L - 1$) has content $e\lambda_r - 1 - r + s$, hence residue i iff $r \equiv b - 1 \pmod{e}$. In particular a single row contributes at most one i -node (as $e \geq 2$ forces $b \not\equiv b - 1$).*

Proof. Immediate from $\text{ct}(r, c) = c - r + s$ and $e \mid e\lambda_r$; the existence conditions for $\mu = e\lambda$ are equivalent to those for λ because $e\lambda_{r-1} > e\lambda_r \Leftrightarrow \lambda_{r-1} > \lambda_r$. $\square$

Lemma 6 (Distinct contents). *Distinct i -nodes of $\mu = e\lambda$ have distinct contents.*

Proof. Two addable nodes in rows $r < r'$ have contents $e\lambda_r - r$ and $e\lambda_{r'} - r'$ (plus s), and $\lambda_r \geq \lambda_{r'}$, $r < r'$ give strict inequality; likewise for two removable nodes. It remains to exclude an addable node in row r and a removable node in row r' with equal content, where by Lemma 5 $r \equiv b$, $r' \equiv b - 1 \pmod{e}$, so $r \neq r'$. If $r' > r$ then $e\lambda_{r'} - 1 - r' < e\lambda_r - r$ since $\lambda_{r'} \leq \lambda_r$ and $r' > r$. If $r' < r$: equality $e\lambda_{r'} - 1 - r' = e\lambda_r - r$ forces $e\lambda_{r'} = e\lambda_r + 1 + r' - r$. Now $r' \equiv r - 1 \pmod{e}$ and $r' < r$, so either $r' = r - 1$, giving $e\lambda_{r-1} = e\lambda_r$, i.e. $\lambda_{r-1} = \lambda_r$ — but then neither node exists, since both require the strict step $\lambda_{r-1} > \lambda_r$; or $r' \leq r - 1 - e$, giving $e\lambda_{r'} \leq e\lambda_r - e$, i.e. $\lambda_{r'} \leq \lambda_r - 1$, contradicting $r' < r$. $\square$

Proposition 7 (Component word). *The i -signature of $\mu = e\lambda$ at charge s , read in increasing content order, is*

$$w = (\text{AR})^n \text{A}^\delta, \quad \delta = \begin{cases} 1 & \text{if } b = 0, \text{ i.e. } s \equiv i \pmod{e}, \\ 0 & \text{otherwise,} \end{cases}$$

for some $n \geq 0$. Moreover $n = 0$ whenever $\lambda = \emptyset$.

Proof. By Lemma 5 the candidate rows for A are $r \equiv b \pmod{e}$ with $0 \leq r \leq L$, and for R they are $r' \equiv b - 1 \pmod{e}$ with $0 \leq r' \leq L - 1$; thus the candidate R-rows are exactly the rows $r - 1$ for candidate A-rows $r \geq 1$. Group them into *blocks* indexed by the candidate A-rows $r \geq 1$, together with the exceptional row $r = 0$ (which occurs iff $b = 0$).

Blocks. Fix a candidate A-row r with $1 \leq r \leq L$.

- If $1 \leq r \leq L - 1$: the addable node at row r exists iff $\lambda_{r-1} > \lambda_r$, and the removable node at row $r - 1$ exists iff $\lambda_{r-1} > \lambda_r$ as well (the condition “ $r - 1 = L - 1$ or $\lambda_r < \lambda_{r-1}$ ” reduces to the same inequality since $r - 1 < L - 1$). So the block contributes either both or neither.
- If $r = L$: the addable node is the new row $(L, 0)$, always present; the removable node at row $L - 1$ is present because $r - 1 = L - 1$. Both occur.

When both occur, their contents are $e\lambda_r - r + s$ (the A) and $e\lambda_{r-1} - r + s$ (the R; here $\lambda_L := 0$), and $\lambda_{r-1} > \lambda_r$ gives A-content $<$ R-content. So each nonempty block contributes the factor AR in increasing content order.

Blocks are content-separated. Two consecutive candidate A-rows are r and $r + e$. The block at $r + e$ occupies contents in $\{e\lambda_{r+e} - r - e, e\lambda_{r+e-1} - r - e\} + s$ and the block at r occupies $\{e\lambda_r - r, e\lambda_{r-1} - r\} + s$. It suffices that $e\lambda_{r+e-1} - r - e < e\lambda_r - r$, i.e. $e\lambda_{r+e-1} < e\lambda_r + e$, i.e. $\lambda_{r+e-1} \leq \lambda_r$, which holds since $r + e - 1 \geq r$. Hence in increasing content order the blocks appear consecutively, deepest row first.

The exceptional row. If $b = 0$ the row $r = 0$ carries an addable i -node at $(0, e\lambda_0)$, always present, of content $e\lambda_0 + s$. This is the largest content among all i -nodes: every A-content is $e\lambda_r - r + s \leq e\lambda_0 + s$ with equality only for $r = 0$, and every R-content is $e\lambda_{r'} - 1 - r' + s < e\lambda_0 + s$. So it is the last letter, giving the trailing A^δ . If $b \neq 0$ row 0 is not a candidate and $\delta = 0$.

The case $\lambda = \emptyset$. Then $L = 0$; candidate A-rows satisfy $0 \leq r \leq 0$ and $r \equiv b$, so only $r = 0$ (and only when $b = 0$), and there are no candidate R-rows. Hence $n = 0$. $\square$

Corollary 8 (Suffix sums of a component word). *With the values $R \mapsto +1$, $A \mapsto -1$, every suffix u of $w = (AR)^n A^\delta$ satisfies*

$$\|u\| \in \begin{cases} \{0, 1\} & \text{if } \delta = 0, \\ \{-1, 0\} & \text{if } \delta = 1. \end{cases}$$

In particular $\max_u \|u\| = 1$ if $\delta = 0$ and $n \geq 1$, and $\max_u \|u\| = 0$ otherwise.

Proof. Read w from the right. For $\delta = 0$ the partial sums of the reversed word alternate $1, 0, 1, 0, \dots$ starting from the empty suffix value 0; for $\delta = 1$ they alternate $-1, 0, -1, 0, \dots$. $\square$

3 The upper bound at level ℓ

Theorem 9 (Individual bound at level ℓ). *Let $e \geq 2$, $\ell \geq 1$, $\mathbf{s} \in \mathbb{Z}^\ell$, $i \in \mathbb{Z}/e\mathbb{Z}$, and let λ be any ℓ -multipartition. Then, for every tie-breaking convention τ in Definition 1,*

$$\varepsilon_i(e\lambda) \leq \#\{j : s_j \not\equiv i \pmod{e} \text{ and } \lambda^{(j)} \neq \emptyset\} \leq \min(|\lambda|, m_i(\mathbf{s})),$$

where $m_i(\mathbf{s}) := \#\{j : s_j \not\equiv i \pmod{e}\}$.

Proof. Write $\mu := e\lambda$ and let Γ be the list of all addable and removable i -nodes of μ in the global reading order of Definition 1; let $w := w_i(\mu)$ be the corresponding word. For a component j let $\Gamma_j \subseteq \Gamma$ be its sublist and w_j the corresponding subword.

Step 1: w_j is the component word. The global order refines increasing content, and by Lemma 6 the nodes of Γ_j have pairwise distinct contents; hence the order induced on Γ_j is exactly increasing content order, and by Proposition 7 $w_j = (AR)^{n_j} A^{\delta_j}$ with $\delta_j = 1$ iff $s_j \equiv i \pmod{e}$, and $n_j = 0$ if $\lambda^{(j)} = \emptyset$.

Step 2: global suffixes restrict to component suffixes. Let u be any suffix of w , cut at a position p . Since the global order restricted to component j is the order of w_j , the letters of u coming from component j form a suffix u_j of w_j . (This is the only property of the reading order we use; it holds for every tie-break τ , and also when nodes from different components share a content.) Hence

$$\|u\| = \sum_{j=1}^{\ell} \|u_j\|.$$

Step 3: conclude. By Lemma 4 and Corollary 8,

$$\varepsilon_i(\mu) = \max_u \|u\| = \max_u \sum_j \|u_j\| \leq \sum_j \max_{u_j} \|u_j\| = \#\{j : \delta_j = 0 \text{ and } n_j \geq 1\}.$$

Finally $\delta_j = 0$ says $s_j \neq i$, and $n_j \geq 1$ forces $\lambda^{(j)} \neq \emptyset$ by Proposition 7; this gives the first inequality. The second follows because the counted set has at most $m_i(\mathbf{s})$ elements and at most $\#\{j : \lambda^{(j)} \neq \emptyset\} \leq |\boldsymbol{\lambda}|$ elements. $\square$

Remark 10 (Where the level-1 argument breaks). *At $\ell = 1$ the right-hand side is ≤ 1 , recovering [5] (“Individual crystal string bound”). The proposed higher-level heuristic was that concatenating reduced component words of shape $A^a R^b$ with $b \leq 1$ would keep a “running maximum” ≤ 1 . Step 2 shows why this fails: ε_i of the merge is the maximum of a sum of ℓ independent per-component suffix sums, not a running maximum of one of them. Nothing prevents ℓ components from attaining their individual maximum 1 at the same cut, and §4 shows they always can.*

Example 11 (The minimal counterexample). Take $e = 2$, $\ell = 2$, $\mathbf{s} = (0, 0)$, $i = 1$, $\boldsymbol{\lambda} = ((1), (1))$, so $e\boldsymbol{\lambda} = ((2), (2))$. Here $b_j = (0 - 1) \bmod 2 = 1$ for both j , so each component has a single block at $r = L = 1$: A at $(1, 0, j)$ of content -1 , R at $(0, 1, j)$ of content $+1$. The global word in increasing content order is

$$w = AARR$$

(the two ties are between letters of equal type, so τ is irrelevant — the *word* here is convention-independent, not merely its reduction). No R precedes an A, so the word is already reduced and $\varepsilon_1(((2), (2))) = 2$. The chain can be checked node by node:

$$((2), (2)) \xrightarrow{\tilde{e}_1} ((1), (2)) \xrightarrow{\tilde{e}_1} ((1), (1)) \xrightarrow{\tilde{e}_1} 0,$$

each arrow removing a genuine 1-node: $((2), (2))$ has a removable 1-node at $(0, 1)$ in each component (content 1); after removing one, $((1), (2))$ has removable nodes of contents 0 and 1, so only the second component offers a 1-node; and $((1), (1))$ has removable nodes of content 0 only, so $\tilde{e}_1$ kills it. Verified for all four tie-break conventions. Thus the bound $\varepsilon_i \leq 1$ is **false** at level 2 already for a 4-box multipartition.

4 Sharpness

Theorem 12 (Sharpness). *Let $e \geq 2$, $\ell \geq 1$, $\mathbf{s} \in \mathbb{Z}^\ell$ and $i \in \mathbb{Z}/e\mathbb{Z}$ with $m_i(\mathbf{s}) \geq 1$. Then there exists an ℓ -multipartition $\boldsymbol{\lambda}$ with*

$$\varepsilon_i(e\boldsymbol{\lambda}) = m_i(\mathbf{s}).$$

An explicit choice is given in the proof; all its components are single columns.

Proof. Set $b_j := (s_j - i) \bmod e$, $c_j := s_j - b_j$ (so $c_j \equiv i \bmod e$), $J := \{j : b_j \neq 0\}$, and note $|J| = m_i(\mathbf{s}) \geq 1$.

Single-column components. Let $\lambda^{(j)} = (1^L)$ for some $L \geq 0$, so $e\lambda^{(j)} = (e^L)$. Since all parts are equal, the block case “ $1 \leq r \leq L - 1$ ” of Proposition 7 never contributes (it needs $\lambda_{r-1} > \lambda_r$). Hence:

- If $b_j \neq 0$ and $L \equiv b_j \pmod{e}$ with $L \geq 1$: the only block is at $r = L$, giving $w_j = AR$ with A-content $= -L + s_j$ and R-content $= e - 1 - (L - 1) + s_j = e - L + s_j$. By Corollary 8 and the proof of Lemma 4, for a cut at content threshold t (i.e. keeping the nodes of content $\geq t$) the component suffix sum is

$$\|u_j\| = 1 \iff t \in (s_j - L, s_j - L + e].$$

Writing $L = b_j + et_j$ with $t_j \geq 0$ gives $s_j - L = c_j - et_j$, so the interval is $(c_j - et_j, c_j - et_j + e]$; as t_j ranges over $\mathbb{Z}_{\geq 0}$ these intervals tile $(-\infty, c_j + e]$.

- If $b_j = 0$ and $L = et$ with $t \geq 1$: the only block is at $r = L$, and there is in addition the trailing $\mathbf{A}$ at row 0 of content $e + s_j$. So $w_j = \mathbf{ARA}$ with contents $s_j - L < s_j + e - L < s_j + e$, and $\|u_j\| = 0$ exactly for $t \in (s_j - et, s_j - et + e]$. For $t = 0$ (i.e. $\lambda^{(j)} = \emptyset$) we have $w_j = \mathbf{A}$ of content s_j and $\|u_j\| = 0$ exactly for $t > s_j$. Together these cover all of $\mathbb{Z}$.

The construction. Put $\theta := \min_{j \in J} c_j$ and let $t^* := \theta + 1$ be the content threshold at which we shall cut.

For $j \in J$ set $t_j := (c_j - \theta)/e$, an integer because $c_j \equiv \theta \equiv i \pmod{e}$, and $t_j \geq 0$ by minimality of θ ; take $\lambda^{(j)} := (1^{b_j + et_j})$. By the first bullet its “ $\|u_j\| = 1$ ” interval is $(c_j - et_j, c_j - et_j + e] = (\theta, \theta + e]$ — the *same* interval for every $j \in J$ — and $t^* = \theta + 1$ lies in it.

For $j \notin J$ set $t_j := \max(0, \lceil (s_j - \theta)/e \rceil)$ and $\lambda^{(j)} := (1^{et_j})$. If $t_j \geq 1$ then $et_j \geq s_j - \theta$ gives $s_j - et_j \leq \theta < t^*$, and $et_j < s_j - \theta + e$ gives $s_j - et_j + e > \theta$, hence $\geq \theta + 1 = t^*$ as these are integers; so t^* lies in the window $(s_j - et_j, s_j - et_j + e]$ on which $\|u_j\| = 0$. If $t_j = 0$ then $s_j \leq \theta < t^*$, and by the second bullet $\lambda^{(j)} = \emptyset$ gives $\|u_j\| = 0$ for every threshold $> s_j$, in particular at t^* .

At the cut determined by t^* we therefore have $\|u_j\| = 1$ for all $j \in J$ and $\|u_j\| = 0$ for $j \notin J$, so by Step 2 of the proof of Theorem 9, $\|u\| = \sum_j \|u_j\| = |J| = m_i(\mathbf{s})$; hence $\varepsilon_i(e\lambda) \geq m_i(\mathbf{s})$ by Lemma 4. The reverse inequality is Theorem 9. $\square$

Corollary 13. *The bound $\varepsilon_i(e\lambda) \leq 1$ is valid for all λ if and only if $m_i(\mathbf{s}) \leq 1$. In particular it fails for every $\ell \geq 2$ as soon as at least two charges s_j are $\not\equiv i \pmod{e}$, and it fails for every $e \geq 2$ and $\ell \geq 2$ at $\mathbf{s} = (0, \dots, 0)$ and $i \neq 0$.*

Corollary 14 (Level-1 consistency). *At $\ell = 1$ we have $m_i(\mathbf{s}) \leq 1$, so Theorem 9 gives $\varepsilon_i(e\lambda) \leq 1$, and Theorem 12 gives equality for $i \not\equiv s_1$; at $i \equiv s_1$ the bound is 0. This reproduces the “Individual crystal string bound” proposition of [5] and its accompanying remark exactly.*

5 Transfer to the Heisenberg descendant

The level-1 theorem C5 concerns not $e\lambda$ itself but the vector $v_{k',e} = P_e^{k'}|\emptyset\rangle$, and the bridge is the expansion $[v_{k',e}] = \sum_{\lambda \vdash k'} f^\lambda[e\lambda]$ [5] (“Expansion of $v_{k',e}$ in the crystal basis at $q = 0$ ”). We record the level- ℓ bridge as a hypothesis, prove that *if* it holds the transfer is automatic, and then report in §6 why we could not test it.

Let B_{-1} be Uglov’s Heisenberg mode at $m = -1$, given on q -wedges by Iijima’s eq. (9) [3, eq. (9)],

$$B_m(u_{\mathbf{k}}) = \sum_{r \geq 1} u_{k_1} \wedge \cdots \wedge u_{k_{r-1}} \wedge u_{k_r - n\ell m} \wedge u_{k_{r+1}} \wedge \cdots,$$

so that B_{-1} shifts one bead up by $n\ell$ ($n = e$), i.e. adds one e -ribbon inside a single component before straightening. Set $v_{k'}^{[\ell]} := B_{-1}^{k'}|\emptyset; \mathbf{s}\rangle$.

Hypothesis 15 ((H2), *not verified here*). $v_{k'}^{[\ell]}$ (or the bar-normalised variant of Remark 16) lies in $\mathcal{L}$ and

$$[v_{k'}^{[\ell]}] = \sum_{|\lambda| = k'} f^\lambda[e\lambda] \pmod{q\mathcal{L}}, \quad f^\lambda = \binom{k'}{|\lambda^{(1)}|, \dots, |\lambda^{(\ell)}|} \prod_j f^{\lambda^{(j)}},$$

the number of standard Young tableaux of the multipartition shape λ .

Remark 16 (A defect in the framing of (H2)). *As stated in the session brief, “Step 1 ($v \in \mathcal{L}$) lifts verbatim” is false as written. At $\ell = 1$ the Leclerc–Thibon closed formula reads $B_{-1}|\lambda\rangle = \sum_{\mu} (-q^{-1})^{h(\mu/\lambda)} |\mu\rangle$, with coefficients in $\mathbb{Z}[q^{-1}]$, not $\mathbb{Z}[q]$; already $B_{-1}^2|\emptyset\rangle$ has q^{-2} coefficients, so*

$v_{k'}^{[1]} \notin \mathcal{L}$. The level-1 theorem is about $P_e|\lambda\rangle = \sum_{\mu}(-q)^h|\mu\rangle$, which is the coefficientwise bar-conjugate of B_{-1} . The correct level- ℓ statement therefore concerns the operator $P_e^{[\ell]}$ whose matrix in the standard basis is obtained from that of B_{-1} by $q \mapsto q^{-1}$ (equivalently, one works in the upper crystal lattice for B_{-1} itself). At $\ell = 1$, $P_e^{[1]} = P_e$ on the nose, so this repairs “ $v_{k'}^{[1]} = v_{k',e}$ up to a correction” into an equality. We verified computationally that our B_{-1} reproduces the $(-q^{-1})^h$ formula exactly at $\ell = 1$ (§8, X1), so the two operators are related exactly as described there.

Proposition 17 (No cancellation). *Suppose $v \in \mathcal{L}$ and $[v] = \sum_{\lambda \in S} c_{\lambda}[e\lambda]$ in $\mathcal{L}/q\mathcal{L}$ with all $c_{\lambda} > 0$ and S finite. Then*

$$\varepsilon_i(v) = \max_{\lambda \in S} \varepsilon_i(e\lambda).$$

Proof. $\tilde{e}_i$ maps each crystal basis element either to 0 or to another crystal basis element, and is injective on the set of basis elements not killed by it (its inverse there is $\tilde{f}_i$). Hence for each n the elements $\tilde{e}_i^n[e\lambda]$, $\lambda \in S$, that are nonzero are pairwise distinct basis elements. Since $\tilde{e}_i^n$ is $\mathbb{Z}$ -linear,

$$\tilde{e}_i^n[v] = \sum_{\lambda \in S} c_{\lambda} \tilde{e}_i^n[e\lambda],$$

a sum of pairwise distinct basis elements with strictly positive coefficients; it vanishes iff every summand does. Therefore $\tilde{e}_i^n[v] \neq 0$ iff $\tilde{e}_i^n[e\lambda] \neq 0$ for some λ , which is the claim. $\square$

Corollary 18 (C5 at level ℓ , conditional on (H2)). *Assume Hypothesis 15. Then for all $k' \geq 0$,*

$$\varepsilon_i(v_{k'}^{[\ell]}) = \max_{|\lambda|=k'} \varepsilon_i(e\lambda) \leq \min(k', m_i(\mathbf{s})),$$

with equality for all k' large enough (explicitly, for $k' \geq |\lambda^|$ where λ^* is the multipartition of Theorem 12, provided λ^* can be enlarged to size k' without decreasing ε_i). In particular, for $\ell \geq 2$, $e = 2$, $\mathbf{s} = (0, 0)$, $i = 1$ and $k' = 2$ one would get $\varepsilon_1(v_2^{[2]}) = 2 > 1$: **the conjecture “ $\varepsilon_i \leq 1$ for all ℓ ” is false, and the correct bound is $m_i(\mathbf{s})$.***

Proof. Combine Hypothesis 15 (all $f^{\lambda} > 0$), Proposition 17, Theorem 9 and Theorem 12. $\square$

Remark 19. *Corollary 18 is stated conditionally and we do not claim it. What is unconditional is Theorems 9 and 12, i.e. everything about the crystal; (H2) is exactly the statement that the q -deformed Heisenberg descendant reduces at $q = 0$ to the multipartitions $e\lambda$, and it is the one thing we could not test.*

6 What blocked (H2): the level- ℓ straightening rule

We report this in detail because it is a reproducible obstruction, not a shortage of time.

To test (H2) one must compute B_{-1} at level ℓ , i.e. apply eq. (9) and then straighten the resulting unordered q -wedges. The straightening rule is Uglov [2, Prop. 3.16], restated as [3, Prop. 5.1]. Writing $k_j = c_j + n(d_j - 1) - n\ell m_j$ with $1 \leq c_j \leq n$, $1 \leq d_j \leq \ell$, the rule reads

$$\begin{aligned} u_{c_2-nm_2}^{(d_2)} \wedge u_{c_1-nm_1}^{(d_1)} &= (-q^{-1})^{\delta_{d_1=d_2}} q^{\alpha} u_{c_1-nm_1}^{(d_1)} \wedge u_{c_2-nm_2}^{(d_2)} \\ &\quad + \operatorname{sgn}(m) (-q^{-1})^{\delta_{d_1=d_2}} (q - q^{-1})^{|m_1-m_2|-\gamma} \sum_{j=\beta} u_{c_2-nm_1-\operatorname{sgn}(m)n_j}^{(d_1)} \wedge u_{c_1-nm_2+\operatorname{sgn}(m)n_j}^{(d_2)}, \end{aligned}$$

with $\alpha, \beta, \gamma, \operatorname{sgn}(m)$ as in [3, Prop. 5.1]. We implemented it verbatim (with $\gamma := 1$ when $d_1 = d_2$, a case the printed definition omits and which is forced by Iijima’s Example 2.5). The implementation:

- reproduces **all four** of Iijima's worked examples exactly: Example 2.1(ii) ($n = 2, \ell = 2$), Example 2.1(iii) ($n = 2, \ell = 2$, a three-fold wedge), and both halves of Example 2.5 ($n = 2, \ell = 3$ and $n = 2, \ell = 1$);
- reproduces the wedge $\leftrightarrow$ (multipartition, multicharge) dictionary of [3, §2.3] exactly, including Example 2.6 (recovering $\mathbf{s} = (2, 0, -2)$, $\boldsymbol{\lambda} = (\emptyset, (1, 1), (4))$ from $\mathbf{k} = (6, 3, 2, 1, -2, -4, \dots)$);
- reproduces, at $\ell = 1$, the Leclerc–Thibon closed formula $B_{-1}|\lambda\rangle = \sum_{\mu} (-q^{-1})^{h(\mu/\lambda)} |\mu\rangle$ on all 79 partitions with $|\lambda| \leq 6$ ($e = 2, 3$) and $|\lambda| \leq 5$ ($e = 4$), with 0 mismatches;
- reproduces the Heisenberg commutation relation $[B_1, B_{-1}] = \frac{1-q^{-2n}}{1-q^{-2}} \cdot \frac{1-q^{2\ell}}{1-q^2}$ [3, §2.5] on $|\emptyset; \mathbf{s}\rangle$ for $(n, \ell, \mathbf{s}) \in \{(2, 1, (0)), (3, 1, (0)), (2, 2, (0, 0)), (3, 2, (0, 0)), (2, 3, (0, 0, 0)), (4, 2, (0, 0))\}$ — i.e. for every *constant* multicharge tested, including the ℓ -dependent factor.

Nevertheless it is **wrong** for $\ell \geq 2$, and we can say exactly how:

1. *Failure of confluence.* The rewriting system is not confluent: straightening always at the leftmost ascent and always at the rightmost ascent give different answers in 244 of 2400 random wedges of length 3–4 over $(n, \ell) \in \{(2, 2), (3, 2), (2, 3), (3, 3), (2, 1)\}$. The smallest instance found is $(n, \ell) = (2, 2)$, $u_{-3} \wedge u_{-1} \wedge u_2$, where the coefficient of $u_1 \wedge u_{-1} \wedge u_{-2}$ comes out as $1 - q^2$ one way and $q^{-2} - 1$ the other. Since $\Lambda^{\mathbf{s}}$ is a quadratic quotient of the tensor algebra, a correct two-factor rule must be confluent, so the rule as we read it is not the correct one.
2. *Non-centrality of the commutator.* Consequently $[B_1, B_{-1}]$ is not a scalar: at $(n, \ell) = (2, 2)$, $\mathbf{s} = (1, 0)$ it acts on $|\emptyset; \mathbf{s}\rangle$ by 4 instead of $q^{-2} + 2 + q^2$ (note $q^{-2} + 2 + q^2 - 4 = (q - q^{-1})^2$, so the defect is exactly one misplaced correction term), and it produces genuine off-diagonal terms such as $\pm(q - q^{-1})[(2), \emptyset]$ from $[(1), (1)]$. The failures are concentrated on the multicharges for which the vacuum bead configuration is *fully packed* (for $n = \ell = 2$ these are exactly $s_1 - s_2 = 1$).
3. *The rule is underdetermined by the local text.* We searched a family of 98,304 readings, allowing: three conditions for α ($c_1 = c_2$, $d_1 = d_2$, both); the lower and upper summation limits β, γ to be *arbitrary* $\{0, 1\}$ -valued functions of $(\text{sgn}(x_1 - x_2), \text{sgn}(m_1 - m_2))$ for $x \in \{c, d\}$ in either assignment; an optional extra q^α on the correction terms; and both d -assignments in the correction terms. Separately we tested all four (c, d) -assignments in the correction terms, of which only the one above matches Iijima's examples. Result: **no reading in the family is simultaneously consistent with Iijima's four examples and confluent**. So the shape of the ansatz — a single leading term plus a j -indexed sum with a uniform $(q - q^{-1})$ coefficient — is itself not what we should be using, and the local text does not determine it.
4. *A structural symptom.* At $q = 1$ all correction terms vanish and the exchange should be that of the classical semi-infinite wedge, i.e. multiplication by -1 . The leading coefficient $(-q^{-1})^\delta q^\alpha$ equals -1 at $q = 1$ when $d_1 = d_2$ but $+1$ when $d_1 \neq d_2$; Iijima's own Example 2.1(ii) exhibits this ($u_{-2} \wedge u_4 = q u_4 \wedge u_{-2} + (q^2 - 1) u_2 \wedge u_0$, giving $u_{-2} \wedge u_4 = u_4 \wedge u_{-2}$ at $q = 1$). We do not know whether this reflects a different sign convention in the identification $\Lambda^{\mathbf{s}}|_{q=1} \cong$ classical wedge, or a transcription problem; it is the natural place to look first.

What is needed: Uglov [2, Prop. 3.16] in the original, which is not available locally. With a correct rule, testing (H2) is a short computation ($\ell = 2$, $e \in \{2, 3\}$, $k' \leq 3$) using the code accompanying this note, in which everything except `swap_rule` is already validated.

7 A dichotomy between the two scaling conventions

The multipartitions produced by the level-1 operator P_e at $q = 0$ have all *parts* divisible by e (height-0 ribbons are horizontal e -strips). Gerber–Norton’s $\mathfrak{sl}_\infty$ -crystal operators, by contrast, act by adding good *vertical* e -strips [4, Thm. 1.8], producing multipartitions in which every part is repeated e times. These two families are transposes of each other, and they behave completely differently:

Observation 20 (computational; not proved here). Let μ be an ℓ -multipartition in which every part of every component is repeated e times. Then $\varepsilon_i(\mu) = 0$ for every i and every $\mathbf{s}$. Verified on 497,814 instances ($e \in \{2, 3, 4, 5\}$, $\ell \in \{2, 3, 4\}$, all $\mathbf{s} \in \{-2, \dots, e+1\}^\ell$, underlying λ of size ≤ 5 , two tie-break conventions), with 0 exceptions.

This is exactly what the bicrystal picture predicts: the $\mathfrak{sl}_\infty$ -crystal commutes with the $\widehat{\mathfrak{sl}}_e$ -crystal, so Heisenberg descendants of the (highest weight) empty multipartition remain $\widehat{\mathfrak{sl}}_e$ -highest weight, i.e. $\varepsilon_i \equiv 0$. Theorems 9–12 say that in the *horizontal* convention — the one the q -deformed operator P_e actually produces at $q = 0$ — highest-weightness fails, and fails by exactly $m_i(\mathbf{s})$. At $\ell = 1$ this defect is the “controlled commutation defect” of [5] (“Structural picture” remark), bounded by 1; the content of the present note is that the defect grows linearly in the number of components whose charge is $\neq i$.

8 Computational verification

All computations are pure Python with exact integer / Laurent-polynomial arithmetic; no CAS. Code in `~/projects/probes/2026-08-29-C5-level-ell/: mpcrystal.py` (level- ℓ crystal), `wedge_level.py` (straightening, B_m , dictionary), `crosscheck.py`, `phase1_individual_bound.py`, `phase1b_sharpness.py`, `phase1c.py`, `phase1d_construction.py`, `findrule.py`, `findrule2.py`.

Check	Range	Instances	Failures
Conventions vs. [5]	worked example + 2 base cases	3	0
Cross-validation vs. <code>probe4_canonical.py</code>	$\ell = 1$, $e \leq 5$, $ \lambda \leq 8$	938	0
Component word $(\text{AR})^n \mathbf{A}^\delta$ (Prop. 7)	(e, ℓ) : 8 pairs, $ \lambda \leq 6$	239 693	0
Bound $\varepsilon_i \leq m_i(\mathbf{s})$ (Thm. 9)	same, $\times 4$ tie-breaks	351 376	0
Refined bound with $\lambda^{(j)} \neq \emptyset$	7 (e, ℓ) pairs, $\times 4$ tie-breaks	663 320	0
Sharpness construction (Thm. 12)	$e \leq 5$, $\ell \leq 4$, all $\mathbf{s}$ in a window	66 464	0
Vertical convention $\varepsilon_i = 0$ (Obs. 20)	$e \leq 5$, $\ell \leq 4$	497 814	0
Iijima Examples 2.1(ii),(iii), 2.5×2 , 2.6	as printed	5	0
X1: $\ell = 1$ reduction of B_{-1} to $(-q^{-1})^h$	$e = 2, 3, 4$	79	0
X2: $[B_1, B_{-1}]$ scalar, constant $\mathbf{s}$	6 settings	6	0
X2: $[B_1, B_{-1}]$ scalar, non-constant $\mathbf{s}$	2 settings	2	2
Confluence of straightening	5 (n, ℓ) pairs, random wedges	2 400	244
Confluence over the search family	98 304 readings	98 304	all

Two entries deserve comment.

- *Independence of the level-1 code.* The level- ℓ crystal in `mpcrystal.py` was written from scratch for this note. Specialised to $\ell = 1$, $\mathbf{s} = (0)$ it agrees with the independently written

and previously validated `probe4_canonical.py` of [5] on all 938 triples (e, λ, i) with $e \leq 5$, $|\lambda| \leq 8$ — not merely on the value of ε_i but on the full ordered list of addable and removable i -nodes.

- *Tie-break dependence.* $\varepsilon_i(e\lambda)$ itself depends on the tie-break τ in 9 802 of 87 844 sampled $(e, \ell, \mathbf{s}, i, \lambda)$; the *bound* of Theorem 9 does not, in any instance. This is the content of Step 2 of its proof and is why we stated the theorem for all τ .
- *Maximum ε_i observed* was exactly ℓ for $(e, \ell) \in \{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (4, 2)\}$ and exactly 1 for $\ell = 1$, matching $m_i(\mathbf{s})$ at $\mathbf{s} = (0, \dots, 0)$.

9 Gaps

We state these precisely.

1. **(H2) is untested.** Hypothesis 15 is neither proved nor refuted. Corollary 18 is conditional on it and is not claimed. The obstruction is located in §6: our reading of the level- ℓ straightening rule is not confluent, and the local text does not determine the correct one. Everything else in the B_{-1} pipeline (eq. (9), the abacus dictionary, the truncation, the $\ell = 1$ specialisation, the constant-charge Heisenberg commutator) is validated.
2. **The framing of $v_{k'}^{[\ell]}$ needs repair** (Remark 16): $B_{-1}^{k'}|\emptyset; \mathbf{s}\rangle$ does not lie in the lower crystal lattice $\mathcal{L}$, so “Step 1 lifts verbatim” is false as written. The right object is the bar-conjugated operator $P_e^{[\ell]}$ (equal to P_e at $\ell = 1$), or equivalently the upper crystal lattice. We have not checked which of the two the correct level- ℓ statement should use; Kwon–Lee’s Props. 5.7 and 5.12 (lower/upper crystal base asymmetry, ψ -twist eq. (5.10)) are the natural place to settle this and were not consulted.
3. **Sharpness in k' is not exact.** Theorem 12 produces a specific λ^* ; we verified $\max_{|\lambda|=k'} \varepsilon_i(e\lambda) \leq \min(k', m_i(\mathbf{s}))$ always (6 597 settings) but equality fails for small k' in 1 746 of them, always with $\max < \min(k', m_i)$. We have not determined the exact threshold $k'_0(e, \ell, \mathbf{s}, i)$ above which equality holds. The construction gives an upper bound for k'_0 (observed $|\lambda^*| \leq 40$ over $e \leq 5, \ell \leq 4$).
4. **Observation 20 is not proved.** It is stated as a computational observation. It is presumably a special case of Gerber–Norton’s bicrystal commutation [4], which we did not read at the depth required to cite.
5. **No claim is made about ε_i for general μ .** Theorems 9 and 12 concern $\mu = e\lambda$ only.

References

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- [2] D. Uglov. *Canonical bases of higher-level q -deformed Fock spaces and Kazhdan–Lusztig polynomials*. In: Physical combinatorics (Kyoto, 1999), Progr. Math., 191:249–299, Birkhäuser, 2000. (Prop. 3.16 — the level- ℓ straightening rule — is the reference we needed and did not have; see §6.)

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